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A.7 Discrete Review

L1 ## Review

- General probability rules -

1. $0 \leq P(A) \leq 1$
2. $P(\text{state space}) = 0$, $P(\phi) = 0$
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
4. If A & B are mutually exclusive,
   then $P(A \cap B) = 0 \Rightarrow P(A \cup B) = P(A) + P(B)$
5. state space $= A \cup A'$, $A \cap A' = \phi$
6. $P(A') = 1 - P(A)$ & vice-versa
7. $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$
8. $P(A \cap B) = P(AB) = P(A) \times P(B|A)$
9. events A & B are independent iff.
   $P(A \cap B) = P(A) \times P(B)$
   then, $P(A|B) = P(A)$

Eg. A & B are independent, $P(A \cup B) = 0.58$,
$P(A) = 0.3$, $P(A' \cup B') = ?$

$P(A \cap B) = x$

$P(A' \cup B') = P((A \cap B)')$

$P((A \cap B)') = 1 - x$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$0.58 = 0.3 + P(B) - 0.3\,P(B)$$
$$0.58 = 0.3 + 0.7\,P(B)$$
$$0.28 = 0.7\,P(B)$$
$$P(B) = 0.4$$

$P(A) \times P(B) = 0.12$

$1 - 0.12 = 0.88$

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Eg. $P(A|B) = 3P(B|A)$, $P(A\cup B) = 7P(A\cap B)$,
$P(B) = 0.1$, $P(A\cap B) = ?$

$$P(A|B) = \frac{P(A)\times \cancel{P(B|A)}}{\cancel{P(A)}\,P(B)} = 3\cancel{P(B|A)}$$

$$P(A) = 3P(B) = 0.3 \quad , P(B) = 0.1$$

$$P(A\cup B) = P(A) + P(B) - P(A\cap B)$$
$$7P(A\cap B) = 0.3 + 0.1 - P(A\cap B)$$
$$8P(A\cap B) = 0.4 \implies P(A\cap B) = \frac{\cancel{0.4}}{\cancel{8}\times 10} = 0.05$$

- Moments of RV

Let X be a RV –

1. y is the mode if,
$$P(X=y) \geq P(X=x) \;\forall x$$
2. m is the median of X if m is the smallest value s.t.,
$$P(X\leq m) = F(m) \geq 1/2$$
$\hookrightarrow$ CDF

3. $E[X] = \sum_x x \cdot P(X=x)$
4. $E[X^2] = \sum_x x^2 \cdot P(X=x)$
5. $Var(X) = E[X^2] - (E[X])^2$
$= E[(X-\mu)^2]$

Eg. $P(X=x) = cx$, $x = 1, 2, \ldots, 5$. Mode? Median?

	1	2	3	4	5
$P(X=x)$	c	2c	3c	4c	5c
CDF	1/15	3/15	6/15	10/15	

5 → mode

4 → median

Eg. $P(X=x) = x/15$ for $x = \{1, \ldots, 5\}$. $Var(X) = ?$

x	$P(X=x)$	$E[X]$	$E[X^2]$
1	1/15	1/15	1/15
2	2/15	4/15	8/15
3	3/15	9/15	27/15
4	4/15	16/15	64/15
5	5/15	25/15	125/15
		Σ 55/15	~~225/15~~ 15

$Var(X) = 15 - 13.3956 = 1.6044$

- Let $A_1, A_2 \ldots A_n$ be a partition of the sample space, i.e., $A_i \cap A_j = \phi$ for $i \neq j$ and $\cup A_i = S$.

$$P(B) = \sum_i P(A_i) \cdot P(B \mid A_i)$$

$$E(X) = \sum_i P(A_i) \cdot E[X \mid A_i]$$

$$P(A_i \mid B) = \frac{P(A_i \cap B)}{P(B)}$$

- $n! = n(n-1)(n-2) \ldots 2 \cdot 1$

  $\hookrightarrow$ no. of ways to order n objects.

  $1! = 1$, $0! = 1$.

- $nC_k = \dfrac{n!}{(n-k)!\,k!} = \dbinom{n}{n-k}$

  $\hookrightarrow$ no. of ways to choose k objects from a set of n.

- Key Distribuⁿ

1. Uniform → all possibilities are equally likely.
2. Bernoulli → 0 or 1
3. Binomial → sum of bernoullis. No. of successes in n indep. trials.
4. Geometric $\{0, 1 \ldots\}$ → no. of failures before first success.
5. Geometric $\{1, 2 \ldots\}$ → no of trials including first success.
6. Negative Binomial → no. of failures before r-th success
   sum of geometrics
7. Poisson → sum of independent poissons is poisson.

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Distribuⁿ	$P(X = x)$	$E[X]$	$Var(X)$
$U\{1 \ldots n\}$	$1/n$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$
Bernoulli (p)	$p,\ x = 1$ $1-p,\ x = 0$	p	$p(1-p)$
Binomial (n, p)	${}^{n}C_{x}\, p^x (1-p)^{n-x}$	np	$np(1-p)$
$G(p)\ \{0, \ldots\}$	$p(1-p)^x$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$
$G(p)\ \{1 \ldots\}$	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	${}^{x+(r-1)}C_{r-1}\, p^r (1-p)^x$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Poisson	$e^{-\lambda} \frac{\lambda^x}{x!}$	λ	λ